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# **Did Remote Work Narrow the Motherhood Penalty?**

**Evidence on the Intensive Margin from Israeli Labor Force  
Survey Microdata, 2017–2023**

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## Abstract

[Harrington and Kahn \(2025\)](#) ask whether the post-pandemic rise of work from home (WFH) reduced the motherhood penalty in the United States. We take the question to Israel and to the intensive margin, using Central Bureau of Statistics (CBS) Labor Force Survey microdata on women aged 25–59 in 2017–2019 and 2021–2023. In difference-in-differences (DiD) and triple-differences (DDD) models of usual weekly hours, with an Israeli-calibrated occupational WFH-exposure index as the third difference, employed mothers’ hours did not change on average (0.23 hours, SE 0.18) but rose where the work can be done from home: the triple interaction is 3.22 hours per unit of exposure (SE 1.02), about 1.8 hours a week between the most and least teleworkable quartiles of occupations, with no gain at zero exposure. The estimate is significant at the 1% level on the analytic standard error, at 5% under a permutation test and at 10% under a wild cluster bootstrap. It survives a saturated fixed-effects specification, a selection correction and the removal of any single occupation, and it is largest for mothers of children under five; fathers show the opposite gradient. The result is about *which* mothers benefit from remote work rather than about mothers as a group.

## 1 Introduction

[Harrington and Kahn \(2025\)](#) pose a simple question for the United States: has the rise of work from home reduced the motherhood penalty in the labor market? Their answer is yes on the extensive margin and on income: a 10% rise in WFH raises mothers’ employment by roughly 0.78 percentage points relative to other women, and employed mothers’ incomes rise by about 1.3%. This paper asks the same question of a different country and a different margin. We use Israeli CBS Labor Force Survey microdata to test whether the post-COVID shift to remote work altered the motherhood penalty in *usual weekly work hours* among employed women. We also estimate the employment margin, but for reasons of statistical power documented in Section 6.8 we treat it as secondary.

The shift to remote work is a rare natural experiment in workplace flexibility, which [Goldin \(2014\)](#) identifies as the last major driver of the remaining gender pay gap. Temporal and locational flexibility is a good mothers pay for, in lower wages or slower promotion, and a shock that lowers the cost of supplying it should be detectable in mothers’ labor supply — for mothers already employed, most directly in how many hours they supply. Unlike the acute disruption of 2020, WFH has proven to be a persistent feature of the post-pandemic labor market ([Barrero et al., 2021](#)), which makes the question policy-relevant rather than merely historical.

The stakes are particularly high in Israel, which combines a productive, internationally integrated labor market with the OECD’s highest total fertility rate, 2.87 children per woman in 2024 against an OECD average of 1.40 ([OECD, 2025](#)): an unusually large

share of the female workforce is at any time the mother of a young child, so an intensive-margin response to flexibility has first-order consequences for aggregate labor input. Israeli practice also diverges from US task-based predictions of which jobs can be done remotely; teaching returned fully to the classroom once schools reopened, whatever its task content says, which matters for how “WFH exposure” should be measured.

Our design exploits the pandemic as the moment at which remote work became a mainstream practice. In the United States, working from home rose from roughly 5% of paid workdays before 2020 to 25–30% afterward (Barrero et al., 2021); in Israel, where the survey began asking only in 2021, between 13% and 16% of employed women aged 25–59 usually worked from home in 2021–2023. We compare mothers and childless women across a pre-period of 2017–2019 and a post-period of 2021–2023, excluding 2020 (Section 3). A difference-in-differences on this contrast estimates the change in the hours penalty. A triple-differences model that interacts it with an occupation’s WFH exposure tests whether remote work is the *mechanism*, by asking whether the change is larger in jobs that can actually be done from home.

Mothers’ hours did not rise on average: the DiD is 0.23 hours (SE 0.18), a reasonably precise null. The effect lives in the interaction with occupational WFH exposure, 3.22 additional weekly hours per unit of exposure, or 1.8 hours between the top and bottom quartiles of occupations, the same order as the 1.74-hour intensive-margin penalty itself, while the implied change at zero exposure is if anything negative. Because the exposure score varies across only forty occupations, we report three inference procedures that place the estimate at the 1%, 5% and 10% levels. It survives a saturated fixed-effects specification, a selection correction, alternative codings of the outcome, the removal of any one occupation, and the exclusion of the occupations whose exposure was calibrated on post-period data; a check on occupational sorting finds no post-2021 movement of mothers toward teleworkable occupations; the gradient is largest for mothers of children under five; and fathers show the opposite gradient. On the employment margin the DiD is a precise zero and the DDD is too imprecise to be informative.

Section 2 places the question in the literature; Section 3 describes the data and the exposure measures; Section 4 shows the hours margin before any model is imposed on it; Section 5 sets out the specifications, inference and selection correction; Section 6 reports the results; Sections 7–8 interpret them; and Section 9 concludes.

## 2 Literature Review

**The motherhood penalty.** The child penalty is one of the most robust facts in labor economics. Kleven et al. (2019b) use Danish administrative data and an event study around the first birth to show that women’s earnings fall by roughly 20% in the long run relative to men’s, through reduced hours, exit from the labor market and moves into

family-friendly, lower-paying jobs; [Kleven et al. \(2019a\)](#) find the pattern universal across countries and its magnitude tied to gender norms more than to family policy. Hours are one of the channels, which is why an intervention that changes the time cost of combining work and care could move the intensive margin even where it does not move employment.

**Temporal flexibility as the mechanism.** [Goldin \(2014\)](#) argues that the last chapter of gender convergence turns on the structure of jobs rather than on human capital. In many occupations firms pay disproportionately for long, contiguous, on-site hours, and workers who cannot supply them, disproportionately mothers, pay a penalty out of proportion to the hours they forgo. Remote work lowers the cost of supplying a given number of hours: it removes the commute, lets hours be split around care obligations, and permits a virtual presence in demanding professions. The prediction for the intensive margin is direct: mothers in occupations where the work can be done from home should supply more hours after the shift, relative to childless women in the same occupations, and mothers in occupations that cannot should not. The ingredients are real: [Mas and Pallais \(2017\)](#) show that workers, and women in particular, give up wages for the option to work from home, and [Cortés and Pan \(2019\)](#) show that the skilled gender gap is concentrated where long hours are rewarded and narrows where substitutes for household time are cheap. Remote work may also cut the other way, through lower visibility and slower advancement, so the sign is an empirical question, and Goldin’s framework says where to look for it: in the gradient of the response across occupations by teleworkability, not in its average.

**Does remote work narrow the gap in practice?** The evidence on whether flexible arrangements realize this potential is mixed. [Bloom et al. \(2015\)](#) find that randomly assigned home working raised work supplied without a fall in output, and [Angelici and Profeta \(2024\)](#) that randomly assigned flexibility shifted the division of household tasks and raised women’s well-being. The acute pandemic period is more ambiguous: [Adams-Prassl et al. \(2020\)](#) show that WFH-feasible jobs were better shielded from the 2020 shock but that women in them absorbed a larger share of the childcare created by school closures (see also [Alon et al., 2020](#)), and for Israel [Yaish et al. \(2021\)](#) document that the 2020 lockdown reallocated paid and unpaid labor within households in gender-unequal ways. This paper uses post-acute (2021–2023) data rather than the initial-shock period most of this literature studies.

**Remote work and hours.** The evidence closest to our outcome concerns what remote work does to hours themselves. [Barrero et al. \(2023\)](#) document that work from home settled at roughly a quarter of paid workdays in the United States after 2021 and that the time it frees is, in workers’ own accounts, split between paid work, care and leisure. Two studies identify the direction of the hours response. [Gibbs et al. \(2023\)](#) follow the

personnel records of information-technology professionals who moved home in 2020 and find total hours up by roughly 18% with output flat, so hours rose even as output per hour fell; [Pabilonia and Vernon \(2022\)](#) use American time diaries and find that on remote days part of the time saved goes to household production and child care rather than to paid work, with the largest reallocation among parents. Those two results bracket the question this paper asks of mothers: whether the time remote work frees is returned to paid hours or absorbed by care. [Emanuel and Harrington \(2024\)](#) show, for call-centre workers, that who selects into remote work differs sharply from what remote work does to them, which is why the design below measures *exposure* from occupational content rather than from realized remote-work status. For Israel, [Budig et al. \(2023\)](#) estimate motherhood penalties in employment and earnings that differ across ethno-religious groups; there is, to our knowledge, no published Israeli estimate of the penalty in hours conditional on working, so the level we report in Section 6 has no direct domestic benchmark.

**Measuring who can work from home.** [Dingel and Neiman \(2020\)](#) classify occupations by whether their tasks can be performed entirely at home and find that 37% of US jobs can. Their classification, mapped to two-digit ISCO-08 codes, is the starting point for our exposure index; because a US task-based measure misclassifies occupations where Israeli practice diverged, we calibrate it against realized Israeli behavior (Section 3.2), for which [Madhala and Bental \(2020\)](#) provide a descriptive baseline. Any occupation-level measure captures only part of the relevant heterogeneity: [Buzaglo-Baris \(2023\)](#) shows that within-firm sorting is a first-order driver of the Israeli gender wage gap.

**Remote work and the motherhood penalty.** The paper closest to ours is [Harrington and Kahn \(2025\)](#), who show for the United States that the rise of WFH raised mothers' employment relative to other women's, driven by fields with high returns to hours and inflexible demands on time, and raised employed mothers' incomes by about 1.3%. That second result is an intensive-margin finding of which hours are one channel; our hours estimate is its closest analogue. We take the question to Israel, to hours conditional on working, with an explicit correction for the selection that conditioning on employment creates, and to an exposure measure calibrated to Israeli practice rather than imported from US task content.

**Identifying a dose-response.** Because the third difference is a continuous exposure score rather than a treated/untreated split, the design belongs to the continuous-treatment difference-in-differences family that [Callaway et al. \(2024\)](#) analyze. Two of their results shape the reading below. Comparing effects *across* exposure levels, which is what a single slope does, requires the stronger condition that occupations at different exposure levels would have followed the same path under any common level of exposure; and the linear

specification returns a weighted average of such comparisons whose weights need not be positive. We therefore report the triple interaction three ways: as the linear slope, under a saturated specification that absorbs every occupation-level margin except the triple interaction, and as quartile contrasts on the bins of the descriptive dose-response figure (Section 5).

## 3 Data and Institutional Context

### 3.1 Source and sample

The data are individual-level microdata from the Israeli Central Bureau of Statistics Labor Force Survey for the survey years 2017–2019 and 2021–2023. The analysis sample is women aged 25–59 (CBS age-group codes 3–7). A woman is a *mother* if she has any child under 17 in the household and childless otherwise; *Post* equals one for survey years 2021 and later. The year 2020 is excluded because its arrangements were emergency improvisation rather than the settled remote-work practice whose effect we want to measure, and because no 2020 CBS extract exists for this project, a constraint that also shapes the exposure index (Section 3.2).

The primary outcome is *usual weekly work hours*. The CBS records usual hours in bins; we assign each bin its median (4, 11, 18, 25.5, 32, 37, 42, 47, 54.5 and 78.5 hours) and impute the two irregular-hours codes from observed medians within the same period. Mid-distribution bins are about five hours wide, so an effect smaller than one bin width, such as the headline estimate below, is best read as a shift in the share of women crossing a bin boundary. Hours are the only intensive-margin outcome the survey supports: the extract records how a wage is paid but not its amount. Hours are defined for women who are employed *and* worked in the survey’s reference week, a restriction that changes the estimand. In the 2017 file, employed respondents absent from work in the reference week had their usual hours recorded as zero, whereas from 2018 onward the same respondents were assigned a real value; left uncorrected, this enters the pre-period as roughly 5,200 spurious zeros, and because absence is more common among mothers (12.4% of employed mothers against 7.3% of childless women) they fall unevenly across the comparison this paper makes. Defining hours on the reference-week workers in *every* year costs about 10% of each year’s employed sample, makes the hours population identical across years, and narrows the estimand to usual hours among women who were working (Section 8).

The secondary outcome is a binary employment indicator, equal to one if the respondent reports being employed and zero otherwise (unemployed and not-in-labor-force are both zero).

All regressions include the same categorical controls: marital status, religion, age group, district of residence and highest educational attainment (six groups). Age enters

as the CBS age-group code because the extract contains no continuous age. CBS survey weights are deliberately not applied in any outcome regression (Section 8).

### 3.2 Constructing WFH exposure

Realized WFH status is itself a labor-market outcome, jointly determined with hours, so the mechanism test needs an occupation-level measure of *exposure* to remote work that the individual does not choose. We build it in three steps, each producing a measure that is reported.

*External.* The [Dingel and Neiman \(2020\)](#) teleworkability score, via an O\*NET/SOC-to-ISCO crosswalk, at the two-digit ISCO-08 level. This is pre-period by construction, but it is a US task-based measure and misclassifies occupations where Israeli practice diverged. Teaching (ISCO 23) is the clearest case: it scores 0.966 on the external index, yet Israeli schools, closed and reopened in stages through 2020 and early 2021 ([Somekh et al., 2021](#); [Hale et al., 2021](#)), were fully in person throughout the 2022–23 survey years, and teachers did not work from home. The crosswalk that maps the US occupation codes onto ISCO-08 is not documented in the replication materials, so a reader can check the forty resulting scores (Appendix Table A1) but not rebuild them from the source.

*Calibrated (primary).* The rule that selects the primary measure is fixed before any estimate is read: the score the headline uses must be anchored on pre-period task content, so that it cannot be built from the post-2021 hours it is meant to explain, and must not place in the treated tail occupations that Israeli institutions kept in person. The external index fails the second condition and the realized index the first; the calibrated index is the only one of the three that meets both, and the other two are reported as bounds in Section 6.4. We correct the external score only where the gap between it and the realized 2022–23 Israeli WFH share is large (greater than 0.5) *and* distinguishable from sampling noise by a one-sided test at the 95% level with a standard error clustered by individual. Ten of the forty two-digit occupations are swapped to their realized values: teaching falls from 0.966 to 0.069, clerical support (ISCO 41) from 1.000 to 0.098, ICT professionals (ISCO 25) from 1.000 to 0.360. Because the swap test draws on 2022–23 data, inside the post-period, the calibrated measure is a documented compromise rather than a fully pre-treatment index; the realized shares are computed on men and women together, and Section 6.4 re-derives them from men alone.

Across the forty occupations, on men and women pooled over 2021–23, the calibrated score has a correlation of 0.53 with the share who worked from home in the reference week and an occupation-size-weighted slope of 0.40 (SE 0.09), so it is relevant. The raw external index fits better in levels (correlation 0.74), because swapping ten occupations onto realized shares while leaving thirty on task-based probabilities mixes two scales. The calibration’s contribution is not a better cross-sectional fit but the removal of ten

grossly mis-scored occupations from the treated tail, and Section 6.4 tests whether that is what drives the result. The forty scores are listed in Appendix Table A1 and plotted in Figure A2.

*Realized.* The realized Israeli WFH share by occupation, anchored to 2021 (no 2020 extract exists) with a minimum of 200 observations per occupation. Post-treatment by construction, it is reported as a bound.

A fourth measure serves three roles. For the employment DDD, occupation is undefined for the non-employed, so that specification uses a pre-period (2017–2019) shift-share index by demographic cell (sex, age group, education, district, marital status, religion and continent of birth; 8,884 cells), equal to the mean of the calibrated occupation score over the cell’s pre-period occupational composition and defined for employed and non-employed women alike. The same index stratifies the selection correction of Section 5.5, and because a woman’s post-2021 occupation cannot move it, it also provides the bound on occupational sorting in Section 6.3. Its dispersion is a third of the occupation-level index’s, so coefficients on the two are compared per standard deviation, never per unit.

## 4 Descriptive Statistics

Nothing in this section is adjusted for the controls of Section 3 or carries an identifying assumption; the point estimates are cell arithmetic, and the intervals around them are clustered by individual because the survey is a rotating panel (Section 5.4).

### 4.1 The sample in summary

Table 1 describes the sample on either side of the treatment/control split. Mothers work *fewer* hours than childless women when they work (38.68 versus 40.06) while being slightly *more* likely to be employed (0.78 versus 0.76), so the level of the penalty is visible on the intensive margin; whether it changed after 2021 is not something the table can show. The two groups are imbalanced in ways the controls must absorb: mothers are the better-educated group (48% with an academic degree against 40%), overwhelmingly married (85% against 47%), and about 0.44 of an age-group code younger, an imbalance that tracks WFH exposure (Appendix Table A2).

### 4.2 The raw hours gap

Mothers work 38.54 hours per week before 2021 and 38.84 after, a gain of 0.31 hours; childless women move from 40.04 to 40.08. The difference between those two changes, 0.27 hours, is the difference-in-differences of equation (1) without a single control, and the regression DiD of Table 2, 0.228 (SE 0.181), is within four hundredths of an hour of it. The controls neither manufacture nor conceal an average effect; its near-absence is

Table 1: Summary statistics: women aged 25–59, by mother status

|                                              | Childless women | Mothers | Difference (SE) |
|----------------------------------------------|-----------------|---------|-----------------|
| <i>Panel A: outcomes</i>                     |                 |         |                 |
| Observations                                 | 133,646         | 239,095 |                 |
| Employment rate                              | 0.7574          | 0.7828  | 0.0254 (0.0031) |
| Usual weekly hours (ref.-week workers), mean | 40.06           | 38.68   | −1.38 (0.09)    |
| standard deviation                           | [12.12]         | [11.62] |                 |
| observations with usual hours                | 93,982          | 164,193 |                 |
| <i>Panel B: characteristics</i>              |                 |         |                 |
| Age-group code (CBS 3–7), mean               | 5.40            | 4.96    | −0.44 (0.01)    |
| Arab (%)                                     | 17.73           | 17.78   | 0.05 (0.33)     |
| <i>Marital status (%)</i>                    |                 |         |                 |
| Married                                      | 46.55           | 84.57   | 38.02 (0.35)    |
| Married, living separately                   | 3.58            | 0.84    | −2.74 (0.11)    |
| Divorced                                     | 13.13           | 7.70    | −5.42 (0.24)    |
| Widowed                                      | 2.75            | 0.99    | −1.76 (0.11)    |
| Single, never married                        | 34.00           | 5.90    | −28.10 (0.31)   |
| <i>Religion of household head (%)</i>        |                 |         |                 |
| Jewish                                       | 73.08           | 79.50   | 6.42 (0.36)     |
| Christian                                    | 7.12            | 2.23    | −4.89 (0.18)    |
| Muslim                                       | 13.78           | 14.89   | 1.11 (0.30)     |
| Druze                                        | 1.88            | 1.64    | −0.24 (0.12)    |
| Other                                        | 4.15            | 1.75    | −2.40 (0.14)    |
| <i>District of residence (%)</i>             |                 |         |                 |
| Jerusalem                                    | 8.76            | 12.31   | 3.56 (0.26)     |
| North                                        | 17.52           | 16.58   | −0.94 (0.32)    |
| Haifa                                        | 14.38           | 11.92   | −2.45 (0.28)    |
| Center                                       | 23.99           | 26.52   | 2.53 (0.36)     |
| Tel Aviv                                     | 21.55           | 14.90   | −6.65 (0.33)    |
| South                                        | 11.66           | 12.37   | 0.72 (0.27)     |
| Judea and Samaria                            | 2.15            | 5.39    | 3.24 (0.15)     |
| <i>Education (%)</i>                         |                 |         |                 |
| Below high school                            | 11.37           | 6.63    | −4.74 (0.23)    |
| High school, no matriculation                | 12.70           | 11.95   | −0.74 (0.24)    |
| Matriculation (Bagrut)                       | 22.73           | 18.82   | −3.90 (0.29)    |
| Post-secondary, non-academic                 | 11.50           | 13.68   | 2.18 (0.24)     |
| Academic degree (BA/MA/PhD)                  | 40.24           | 47.97   | 7.73 (0.39)     |
| Other / no certificate                       | 1.47            | 0.95    | −0.52 (0.07)    |

Israeli CBS Labor Force Survey, survey years 2017–2019 and 2021–2023, women aged 25–59; a mother has any child under 17 in the household. Weekly hours are conditional on working in the reference week, with the standard deviation in brackets and their own observation count. The difference column is mothers minus childless women, with a standard error clustered by individual. All figures are *unweighted*. The Arab share is the CBS population-group code used for the subgroup analysis; religion is a separate item recorded for the household head. Marital status, religion, district, age group and education are the five controls in every regression.

already in the arithmetic of four averages. Mothers remain well below childless women in both periods.

### 4.3 Hours year by year

Figure 1 unpools the pre-period. The raw mother-minus-non-mother gap is  $-1.37$  hours in 2017,  $-1.48$  in 2018 and  $-1.67$  in 2019, three years within a third of an hour of each other and each within the others' confidence bounds: the groups were not diverging before treatment. The post-period path is  $-1.37$ ,  $-1.53$  and  $-0.80$ , not monotone and not individually distinguishable, so the timing reading of Section 7 rests on the event studies of Section 6, not on this series.

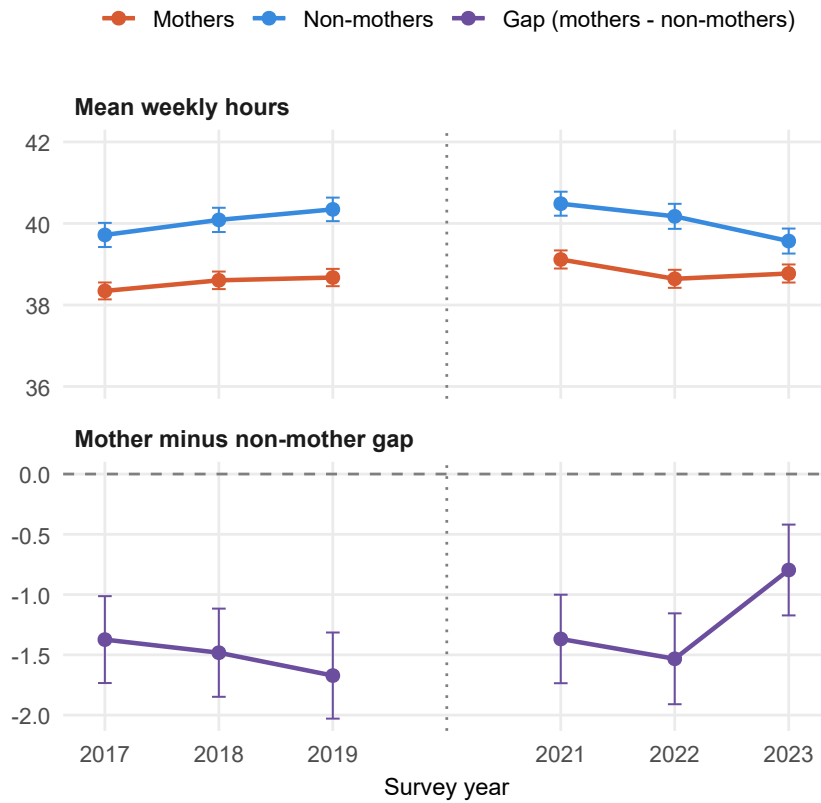


Figure 1: Weekly hours by survey year: levels and the motherhood gap  
*Notes:* Women aged 25–59 who worked in the reference week, unweighted means with no controls. The upper panel plots each group's mean weekly hours on an axis spanning 36 to 42 hours, so year-to-year movements of under half an hour should be read against that range; the lower panel plots their difference. Vertical bars are 95% confidence intervals clustered by individual. The dotted rule marks the excluded 2020 survey year, and the series break there rather than interpolating across it. The three pre-period gaps are the raw counterpart of the insignificant  $\text{Mother} \times \text{year}$  coefficients in Figure 3.

## 4.4 Exposure and the commuting channel

Figure 2 computes the same difference-in-differences separately within each quartile of occupational WFH exposure. The bottom three quartiles return  $-0.32$ ,  $0.27$  and  $-0.02$  hours, none distinguishable from zero; the top quartile returns  $1.55$  hours (95% CI  $[0.74, 2.36]$ ). The near-zero average of Section 4.2 is therefore a weighted average of three nulls and one substantial positive difference, and the dose-response the DDD summarises as a single slope is visible without any regression. It is not linear in exposure: the bottom three quartiles span exposure values of  $0.00$  to  $0.30$  and the top quartile  $0.30$  to  $0.75$ , so most of the regressor's dispersion, and the identifying contrast, sits in one tail.

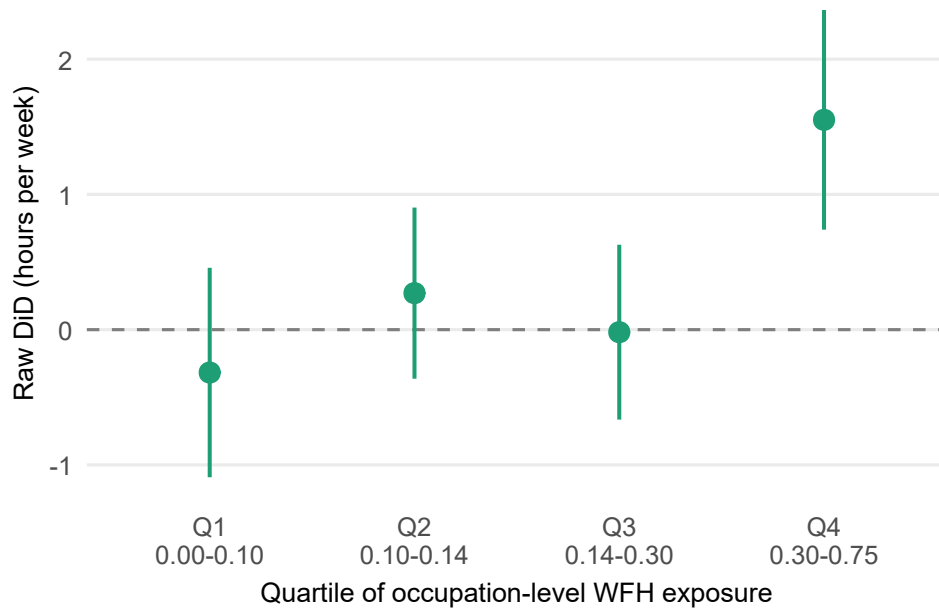


Figure 2: Raw hours difference-in-differences by quartile of WFH exposure

*Notes:* Women aged 25–59 who worked in the reference week and carry a matched occupation code. Each point is the raw difference-in-differences within one quartile of the occupation-level calibrated exposure measure, from unweighted cell means with no controls; bars are 95% confidence intervals clustered by individual. Breakpoints are computed on pre-period rows and applied to all years; each quartile's exposure range is printed beneath its tick.

Commuting patterns are consistent with this channel. Mothers were the *more* mobile group in every year, working outside their locality of residence 48.2% of the time in 2019 against 46.1% of childless women, and their commuting fell less in 2021 (1.5 against 3.9 percentage points, a difference of 2.39 points, SE 1.17). That fits remote work relaxing a time constraint for women who were already commuting rather than drawing a home-bound group into work; it is descriptive evidence, and no specification uses locality of work as an outcome.

## 5 Empirical Strategy

### 5.1 Hypothesis

The flexibility framework of [Goldin \(2014\)](#) yields one testable hypothesis. **H1:** after 2021, mothers' usual hours conditional on working rise relative to childless women's in proportion to their occupation's WFH exposure, with no response where the work cannot be done from home. The exposure gradient, not the average change, is the object of interest.

### 5.2 Difference-in-differences

For outcome  $Y_{it}$  (weekly hours conditional on working that week, or employment) we estimate

$$Y_{it} = \beta_0 + \beta_1 \text{Mother}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Mother}_i \times \text{Post}_t) + X'_{it} \gamma + \varepsilon_{it}, \quad (1)$$

where  $X_{it}$  is the control vector described above and  $\beta_3$  is the differential change in the outcome for mothers relative to childless women after the shift to remote work.

### 5.3 Triple differences

To test whether the narrowing operates through remote work, we interact the DiD with occupational WFH exposure:

$$\begin{aligned} Y_{it} = & \alpha + \delta_1 \text{Mother}_i + \delta_2 \text{Post}_t + \delta_3 \text{WFH\_Exposure}_{j(i)} + \delta_4 (\text{Mother}_i \times \text{Post}_t) \\ & + \delta_5 (\text{Mother}_i \times \text{WFH\_Exposure}_{j(i)}) + \delta_6 (\text{Post}_t \times \text{WFH\_Exposure}_{j(i)}) \\ & + \delta_7 (\text{Mother}_i \times \text{Post}_t \times \text{WFH\_Exposure}_{j(i)}) + X'_{it} \gamma + \varepsilon_{it}, \end{aligned} \quad (2)$$

where  $j(i)$  is individual  $i$ 's occupation.  $\delta_7$  asks whether the post-shift change for mothers relative to non-mothers is larger in more exposed occupations.  $\delta_6$  absorbs any post-2021 change in teleworkable occupations common to mothers and childless women, the general remote-work recovery;  $\delta_5$  absorbs the pre-existing level of the penalty across occupations, so that  $\delta_7$  is a change and not a level; and  $\delta_4$  is the post-shift change for mothers at *zero* exposure, not an average effect. Following [Olden and Møen \(2022\)](#), the DDD requires a single parallel-trends assumption: that the mother/non-mother gap in high- versus low-exposure jobs would have evolved in parallel absent the shift to remote work.

For the hours DDD,  $\text{WFH\_Exposure}_j$  is the calibrated occupation-level score and the sample is employed women matched to a two-digit occupation; for the employment DDD,  $\text{WFH\_Exposure}$  is the cell-based index, the sample is all women, and a  $\text{Mother} \times \text{age-group}$  interaction is added for the pre-period age imbalance.

Equation (2) makes each lower-order occupation-level difference linear in exposure, an unnecessary restriction with forty occupations, so we estimate two relaxations alongside it. The *saturated* specification replaces every lower-order term with fixed effects,

$$Y_{it} = \delta_7(\text{Mother}_i \times \text{Post}_t \times \text{WFH\_Exposure}_{j(i)}) + X'_{it}\gamma + \lambda_{j(i),t} + \mu_{j(i),\text{Mother}_i} + \nu_{\text{Mother}_i,t} + \varepsilon_{it}, \quad (3)$$

where  $\lambda_{jt}$ ,  $\mu_{j,m}$  and  $\nu_{m,t}$  are occupation-by-year, occupation-by-mother and mother-by-year effects; they absorb, respectively,  $\delta_2$ ,  $\delta_3$ ,  $\delta_6$  and any occupation-specific year shock;  $\delta_1$  and  $\delta_5$ ; and  $\delta_4$ . The *binned* specification replaces  $\text{WFH\_Exposure}_{j(i)}$  with indicators for the pre-period exposure quartile the occupation falls in, the bins of Figure 2, so that the top-versus-bottom contrast carries a standard error. For each DDD we also report the minimum detectable effect on the reference distribution the model’s inference uses,  $\text{MDE} = \text{SE}(t_{0.975} + t_{0.80})$ , which is  $2.87 \times \text{SE}$  with 39 degrees of freedom and  $2.80 \times \text{SE}$  under the normal, so that a null can be read as “genuinely null” or “underpowered.”

## 5.4 Inference

The survey is a rotating panel. The 258,175 reference-week workers of Table 1 come from 64,602 distinct respondents, about four observations each and two to three within a single survey year, with 96% of rows belonging to someone who appears more than once; the regressions use 251,857 of those rows from 63,199 respondents, the difference being the 6,318 rows whose education code is missing (education is the only control with missing values). Treating the observations as independent would understate standard errors by a factor of roughly 1.8, so equation (1) and every descriptive interval in Section 4 are clustered by individual. What the panel structure does *not* support is within-person identification: of the 80,560 women in the full sample, 1.9% are interviewed on both sides of the pre/post boundary, so an individual fixed-effects design was considered and rejected.

For the hours DDD the regressor is constant within a two-digit occupation, and standard errors are clustered at that level, leaving forty clusters. The reported  $p$ -values use a  $t$  reference distribution with  $G - 1 = 39$  degrees of freedom and a cluster-count adjustment to the variance. Forty is few enough that we do not rest on that approximation: for the headline coefficient and for the pre-period coefficients of the triple-difference event study we also report wild cluster bootstrap  $p$ -values, with Rademacher weights, the null imposed and 9,999 draws (Roodman et al., 2019; Fischer and Roodman, 2021), and Section 5.6 adds a permutation test that relies on no asymptotic approximation at all. The joint pre-period test remains the analytic Wald  $F$ , since the joint bootstrap is not available in that implementation without a Julia toolchain. The employment DDD is clustered on the age-by-education-by-district cell (about 210 clusters).

## 5.5 Selection into employment: Lee bounds

Hours are observed only for women who are working, and whether a woman is working is itself an outcome of the same treatment: if remote work pulls marginal mothers into work after 2021, the post-period sample of working mothers differs from the pre-period one for reasons unrelated to hours choices. We address this with trimming bounds in the spirit of Lee (2009), adapted to the two-by-two design. Let  $s_{ab} = \Pr(\text{hours observed} \mid \text{Mother} = a, \text{Post} = b)$ , the probability of being employed *and* working in the reference week. Under parallel trends in selection the counterfactual for post-period mothers is  $s_{11}^* = s_{10} + (s_{01} - s_{00})$ ; if the observed  $s_{11}$  exceeds it, the excess share  $1 - s_{11}^*/s_{11}$  is trimmed from the (Mother = 1, Post = 1) cell’s hours distribution, from the top for a lower bound and from the bottom for an upper bound, and equation (1) is re-estimated on each trimmed sample. Trimming from a single tail assumes monotone selection. For the DDD the counterfactual is computed within each quartile of the cell-based exposure index, which unlike the occupation-level index is observable for the non-employed; each quartile’s post-period mothers are trimmed by their own excess share, the trimmed subsets are recombined, and equation (2) is refit with the occupation-level regressor. This assumes selection depends on exposure only through the cell-index quartile. Because the parameter is partially identified we also report the Imbens and Manski (2004) confidence interval for the identified set. The construction handles only *excess* selection in the treated cell (Section 8).

## 5.6 Diagnostics, falsification, and the fathers’ comparison

We estimate event-study versions of equation (1) with Mother  $\times$  year interactions (reference year 2019) for both outcomes and test the pre-period coefficients jointly. Because those test the difference-in-differences assumption rather than the triple-differences one, we also estimate the event-study version of equation (2) itself, with Post replaced by survey-year interactions at every level, and apply the same joint test to its pre-period coefficients.

The falsification test is a permutation test on the exposure score. Under the sharp null that teleworkability is unrelated to the post-2021 change in the motherhood gap, the forty occupation scores are exchangeable across occupations, so we reassign them at random across the forty occupation codes 999 times, holding every woman’s occupation and the mother/period structure fixed, refit equation (2), and record the cluster-robust  $t$ -statistic of the triple interaction. The permutation  $p$ -value is the share of draws, counting the observed one, whose  $|t|$  reaches the observed value (Phipson and Smyth, 2010). It needs neither the  $t(39)$  approximation nor a bootstrap, and it answers the question a placebo should: how often would a design like this one produce an exposure gradient as large as ours when exposure is, by construction, noise.

Occupation is observed when the outcome is, so it is itself a potential outcome: if remote work drew mothers into teleworkable occupations after 2021, the post-period composition of mothers inside high-exposure occupations changes for reasons unrelated to hours. We check this directly with a difference-in-differences that takes the exposure score itself as the outcome, and bound it by re-estimating the hours DDD on the pre-period cell index of Section 3.2, which a woman’s later occupation cannot move.

The remaining checks re-estimate the hours DDD with an added Mother  $\times$  age-group interaction and with pre-period age-group raking weights; with the external and realized indices in place of the calibrated one, with the index calibrated on men only, and with the ten swapped occupations given their own Mother  $\times$  Post terms; with each occupation removed in turn; with alternative codings of the outcome; separately for Jewish and Arab women; by the age of the youngest child; and on the male subsample with “mother” read as “father.” The last is a comparison, not a placebo: fathers in teleworkable occupations are exposed to remote work exactly as mothers are, so a negative gradient would be evidence on household reallocation rather than on a confound. Coefficients across disjoint subsamples are compared with two-sample  $z$ -tests.

## 6 Results

Throughout, significance is denoted \*\*\*  $p < 0.001$ , \*\*  $p < 0.01$ , \*  $p < 0.05$ , ·  $p < 0.1$ . Standard errors are in parentheses.

### 6.1 Pre-trends and the hours DiD

Figure 3 reports the hours event study. Relative to 2019, neither pre-period Mother  $\times$  year coefficient is distinguishable from zero (0.352 in 2017 and 0.201 in 2018) and a joint Wald test on the two does not reject ( $F(2, 63, 198) = 0.949$ ,  $p = 0.387$ ); the same test on the employment outcome also passes (Section 6.8). The triple difference rests on a stricter assumption and has an event study of its own (Section 6.3). In the post-period the coefficients are 0.392 in 2021, 0.121 in 2022 and 0.752\*\* in 2023.

Column (1) of Table 2 reports equation (1). Mothers work 1.739 fewer hours per week than childless women, the intensive-margin penalty itself, and that penalty did not measurably change after the shift to remote work: Mother  $\times$  Post = 0.228 (SE 0.181,  $N = 251,857$ ). The unclustered standard error would be 0.098, roughly half the clustered one, which is why the raw arithmetic of Section 4.2 should not be read as a significant increase. The minimum detectable effect is about 0.51 hours, so the estimate rules out average gains much above half an hour a week: a reasonably precise null. Survey-year effects in place of the single Post dummy leave it at 0.231 (SE 0.181). Selection into the estimation sample does not change the reading: post-period mothers are observed

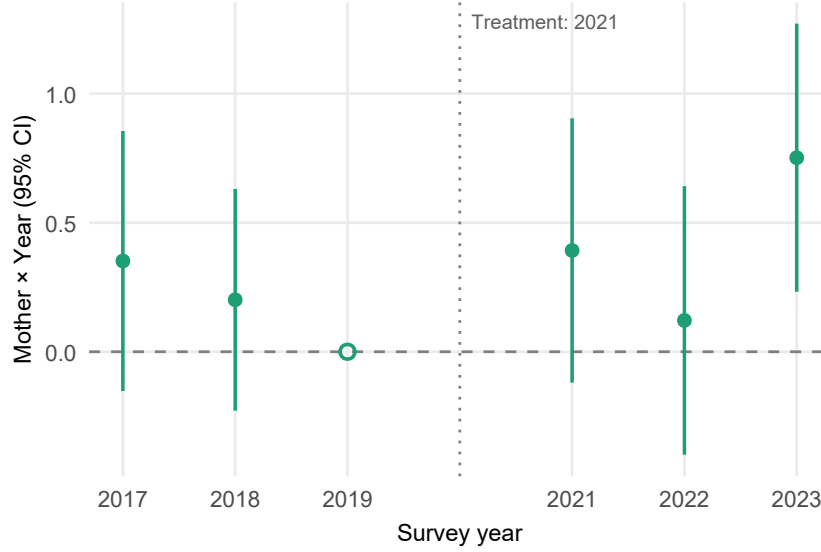


Figure 3: Event study: Mother  $\times$  year interactions for weekly hours, reference year 2019

*Notes:* Women working in the reference week ( $N = 251,857$ ); standard controls; standard errors clustered by individual. Vertical bars are 95% confidence intervals. Coefficients (SE): 2017, 0.352 (0.257); 2018, 0.201 (0.219); 2021, 0.392 (0.261); 2022, 0.121 (0.265); 2023, 0.752\*\* (0.265). Joint test on 2017–2018:  $F(2, 63, 198) = 0.949$ ,  $p = 0.387$ . The hollow point at 2019 is the omitted reference year. The gap between 2019 and 2021 is the excluded 2020 survey year; the dotted rule marks the treatment boundary.

with hours slightly more often than the parallel-trends-in-selection counterfactual implies ( $s_{11} = 0.696$  against 0.683), the Lee bounds after trimming the 1.9% excess are  $-0.517$  (SE 0.178) and 0.802 (SE 0.179), and the Imbens–Manski 95% interval for the identified set is  $[-0.810, 1.096]$ . Nothing that follows rests on the plain DiD.

## 6.2 The hours DDD: the paper’s headline result

Column (2) of Table 2 interacts the DiD with the calibrated occupation-level exposure index. The triple interaction is 3.224\*\* (SE 1.022,  $N = 246,326$ ): among working mothers, those in more WFH-exposed occupations saw a larger post-2021 increase in weekly hours relative to childless women in the same occupations. The index has a standard deviation of 0.199, so the estimate is 0.64 hours per standard deviation, and between the quartiles of Figure 2, whose mean exposures are 0.06 and 0.59, the regression implies 1.71 hours a week against 1.87 from the raw cell means. At zero exposure the implied change for mothers is  $-0.539$  (SE 0.336), and because  $\text{Post} \times \text{WFH\_Exposure}$  is in the model the gradient is net of whatever happened to hours in teleworkable occupations for all women: outside them there is no gain to speak of. The estimate is 3.15 analytic standard errors from zero, qualified by the bootstrap and permutation  $p$ -values of Section 6.4; the minimum detectable effect on the  $t_{39}$  reference is 2.94 hours per unit, or 0.58 per standard deviation,

so an effect of this size is detectable without much room to spare.

Columns (3) and (4) relax the functional form. With occupation-by-year, occupation-by-mother and mother-by-year fixed effects in place of every lower-order term (equation (3)), the triple interaction is 2.876 (SE 0.950), a tenth below the pooled estimate and still significant at the 1% level on the analytic standard error. Column (4) replaces the score with indicators for the pre-period exposure quartile: relative to the least teleworkable quartile, the post-2021 change in the gap is 0.753 hours (SE 0.483) in the second quartile, 0.600 (SE 0.485) in the third and 1.844 (SE 0.595,  $p < 0.01$ ) in the top quartile, the only contrast that is individually significant. The response is a step at the top of the exposure distribution more than a smooth gradient, and that top quartile holds six of the forty occupations, which is why it matters that the pooled and saturated columns, which use all forty, agree with it in sign and magnitude, and that no single occupation drives it (Section 6.4).

### 6.3 Diagnostics for the triple difference

**Pre-trends.** The pre-trend test of Figure 3 licenses equation (1), not equation (2): the triple difference requires that the mother/non-mother gap trended together *across levels of WFH exposure*, and a pre-period divergence correlated with exposure but netting to zero across occupations would pass the DiD test while invalidating the headline. Figure 4 therefore reports the event-study analogue of equation (2), on the same sample, index and clustering as column (2) of Table 2. Neither pre-period coefficient is close to significant or far from zero (0.587 in 2017 and 0.313 in 2018), the joint Wald test does not come near rejecting ( $F(2, 39) = 0.041$ ,  $p = 0.960$ ), and the post-period path steps up in 2021 and rises: 3.011\*, 3.545\*\*, 4.020\*\*, all straddling the pooled estimate. The test has limited power: with forty clusters the pre-period standard errors (2.24 and 1.74) are of the same order as the effect, so each pre-period interval contains values as large as the estimate itself, and the wild cluster bootstrap ( $p = 0.84$ , 0.88 and 0.85 for the two coefficients and their sum) does not change that. The test shows no sign of a violation; it does not rule one out.

**Lee bounds.** Table 3 applies the generalized selection correction. Excess selection among post-period mothers is present in all four quartiles of the cell-based index but modest: the trim proportion ranges from 0.80% to 4.48%, and 1,396 of 72,820 post-period working mothers are trimmed. The bounds are 3.126 (SE 1.144) and 3.443 (SE 1.022), and the Imbens–Manski 95% confidence interval for the identified set, [1.021, 5.324], excludes zero.

**Sorting across occupations.** The bounds handle selection into the estimation sample, not selection across occupations within it. If remote work drew mothers into teleworkable

Table 2: Intensive-margin DiD and DDD: weekly hours, reference-week workers

|                                            | (1) DiD              | (2) DDD              | (3) DDD, saturated | (4) DDD, quartiles   |
|--------------------------------------------|----------------------|----------------------|--------------------|----------------------|
| Mother $\times$ Post $\times$ WFH_Exposure |                      | 3.224**<br>(1.022)   | 2.876**<br>(0.950) |                      |
| Mother $\times$ Post $\times$ Q2 (vs. Q1)  |                      |                      |                    | 0.753<br>(0.483)     |
| Mother $\times$ Post $\times$ Q3 (vs. Q1)  |                      |                      |                    | 0.600<br>(0.485)     |
| Mother $\times$ Post $\times$ Q4 (vs. Q1)  |                      |                      |                    | 1.844**<br>(0.595)   |
| Mother $\times$ Post                       | 0.228<br>(0.181)     | -0.539<br>(0.336)    |                    | -0.472<br>(0.346)    |
| Mother $\times$ WFH_Exposure               |                      | -1.295<br>(1.053)    |                    |                      |
| Post $\times$ WFH_Exposure                 |                      | -0.548<br>(0.810)    |                    |                      |
| WFH_Exposure                               |                      | 1.128<br>(3.912)     |                    |                      |
| Mother                                     | -1.739***<br>(0.144) | -1.449***<br>(0.351) |                    | -1.527**<br>(0.452)  |
| Post                                       | -0.057<br>(0.148)    | 0.053<br>(0.273)     |                    | -0.130<br>(0.367)    |
| Constant                                   | 32.355***<br>(0.318) | 32.115***<br>(1.540) |                    | 31.581***<br>(2.051) |
| MDE for triple interaction ( $t_{39}$ )    |                      | 2.938                |                    |                      |
| SD of exposure, estimation sample          |                      | 0.199                |                    |                      |
| Triple interaction per SD of exposure      |                      | 0.640                |                    |                      |
| Occupations in Q1 / Q2 / Q3 / Q4           |                      |                      |                    | 16 / 7 / 11 / 6      |
| Fixed effects                              | None                 | None                 | Three sets         | None                 |
| Observations                               | 251,857              | 246,326              | 246,324            | 246,326              |
| $R^2$                                      | 0.040                | 0.041                | 0.135              | 0.048                |
| Clustering                                 | individual           | occupation (40)      | occupation (40)    | occupation (40)      |

Dependent variable: usual weekly work hours, women working in the reference week; standard controls in every column. Columns (2)–(4) use the calibrated occupation-level exposure index on column (1)’s sample less the rows with no matched two-digit occupation. Column (3) is equation (3), whose three sets of fixed effects (occupation  $\times$  year, occupation  $\times$  mother, mother  $\times$  year) absorb every lower-order term. Column (4) replaces the score with indicators for its pre-period quartile (Figure 2’s bins); its lower-order quartile terms are estimated but not shown, and the number of occupations per bin is listed because occupation is the cluster. The MDE uses the  $t_{39}$  reference the column’s inference uses ( $\alpha = 0.05$ , power 0.80, multiplier 2.87). Columns (2)–(4) use a  $t$  reference distribution with 39 degrees of freedom (Section 5.4); the wild-bootstrap and permutation  $p$ -values for column (2)’s triple interaction are in Table 4. Column (2)’s Mother  $\times$  Post is the effect at *zero* exposure, not an average effect. Coefficients and standard errors to three decimals throughout. No coefficient is dropped for collinearity in any column.

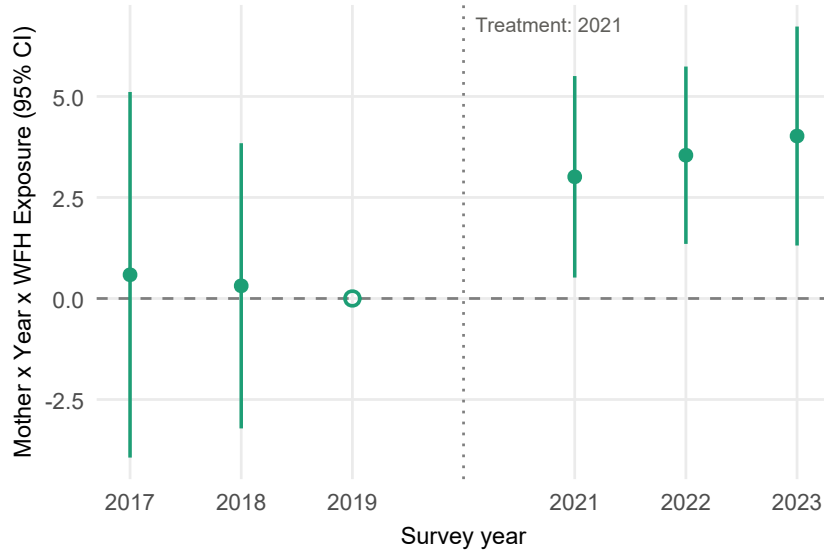


Figure 4: Event study for the triple difference: Mother  $\times$  year  $\times$  WFH\_Exposure interactions for weekly hours, reference year 2019

*Notes:* Sample, controls, exposure and clustering as in column (2) of Table 2 ( $N = 246,326$ , 40 clusters). Vertical bars are 95% confidence intervals. Coefficients (SE), hours per unit of exposure: 2017, 0.587 (2.237); 2018, 0.313 (1.745); 2021, 3.011\* (1.233); 2022, 3.545\*\* (1.085); 2023, 4.020\*\* (1.340). Joint test on 2017–2018:  $F(2, 39) = 0.041$ ,  $p = 0.960$ ; wild cluster bootstrap  $p$ -values (Rademacher, 9,999 draws) 0.838 for 2017, 0.884 for 2018 and 0.852 for their sum. The specification includes the year main effects and both two-way year interactions alongside the triple interaction plotted here. The hollow point at 2019 is the omitted reference year; the gap between 2019 and 2021 is the excluded 2020 survey year.

occupations after 2021, the exposure score itself, as an outcome, would show it. It does not: on the DDD’s own sample, a difference-in-differences with the occupation-level score as the dependent variable gives  $\text{Mother} \times \text{Post} = 0.0013$  (SE 0.0035), under one hundredth of the score’s standard deviation, the probability of being in the top quartile moves by 0.4 percentage points (SE 0.7), and the event-study version is flat in every year (Table 4, *Occupational sorting*). The bound that does not depend on the current occupation, the hours DDD on the pre-period demographic-cell index, is 0.139 hours per standard deviation of that index (SE 0.159, 245 clusters) against 0.640 per standard deviation of the occupation score; the cell index is a far coarser measure of a woman’s own teleworkability and attenuates toward zero, so the bound is positive and imprecise, and the sorting DiD is the informative test.

*Panel A: selection rates and trimming by cell-exposure quartile*

Table 3: Generalized Lee bounds on the hours DDD ( $\text{Mother} \times \text{Post} \times \text{WFH\_Exposure}$ )

| Quartile | $s_{00}$ | $s_{01}$ | $s_{10}$ | $s_{11}$ | $s_{11}^*$ | $n(\text{Mother}=1, \text{Post}=1)$ | Trim  |
|----------|----------|----------|----------|----------|------------|-------------------------------------|-------|
| 1        | 0.5830   | 0.5862   | 0.5497   | 0.5788   | 0.5529     | 21,655                              | 4.48% |
| 2        | 0.7070   | 0.7120   | 0.6856   | 0.6977   | 0.6906     | 28,540                              | 1.01% |
| 3        | 0.7542   | 0.7450   | 0.7554   | 0.7522   | 0.7462     | 34,293                              | 0.80% |
| 4        | 0.8194   | 0.8029   | 0.7352   | 0.7399   | 0.7188     | 21,815                              | 2.85% |

*Panel B: bounds on the triple interaction*

| Bound                                       | Coefficient | SE    | 95% CI         |
|---------------------------------------------|-------------|-------|----------------|
| Lower                                       | 3.126       | 1.144 | [0.883, 5.368] |
| Point (untrimmed)                           | 3.232       | 1.054 | [1.166, 5.297] |
| Upper                                       | 3.443       | 1.022 | [1.440, 5.446] |
| Imbens–Manski 95% CI for the identified set |             |       | [1.021, 5.324] |

$s_{ab}$  is the probability that hours are observed (employed *and* working in the reference week). Panel A’s  $n$  column counts all post-period mothers in the quartile, working or not; the trim proportions apply to the occupation-matched subset with observed hours (1,396 of 72,820 in total), the population the bound regressions use. The untrimmed point estimate here (3.232) differs slightly from Table 2 (3.224) because this sample additionally requires a matched cell-based exposure. Outcome regressions use the occupation-level index and cluster by occupation.

## 6.4 Robustness

Table 4 collects the remaining checks.

**Exposure measure.** With the raw [Dingel and Neiman \(2020\)](#) external index in place of the calibrated one the triple interaction is 0.672 (SE 0.884), positive but insignificant;

with the realized 2021-anchored index it is 5.195\* (SE 2.363). The external index is the only one built entirely from pre-treatment information and it yields the weakest estimate, so the natural objection is that calibrating on 2022–23 realized WFH lets the regressor absorb post-period outcomes. Three checks address it while holding the sample at all forty occupations. Calibrating on men only, so that no woman in the regression contributes to her own regressor, swaps twelve occupations rather than ten and gives 3.869\*\*\* (SE 0.979). Keeping the external score as the regressor but letting the ten swapped occupations carry their own Mother  $\times$  Post change gives an external-score gradient of 2.339 (SE 1.352,  $p = 0.09$ ) beside a negative change for the swapped group ( $-1.347$ , SE 0.926): the occupations the external index places in the treated tail did not respond, and once they are allowed not to, the gradient in the external score re-emerges at two thirds of the headline. Dropping them altogether, so that the two indices coincide on the thirty occupations that remain, gives 4.309\*\* (SE 1.192, 30 clusters). The calibration does not produce the result; what it does is remove from the treated tail occupations, teaching and clerical work above all, that returned to the workplace whatever their task content. What it cannot be made into is a pre-treatment measure, and a reader who weights the external index above the others will read the headline as directionally supported rather than established.

**Covariate balance and age.** Appendix Table A2 reports the pre-period difference between mothers and childless women in every control within each exposure quartile of the estimation sample, and the differences vary with exposure: mothers are younger than childless women by 0.54 of an age-group code in the least teleworkable quartile and older by 0.08 in the most, the academic-degree gap falls from 10.8 percentage points to under 2, and the marriage gap rises from 32 to 43 points. A gap that itself varies with exposure is what the triple interaction is identified off, so two specifications relax the age control. A Mother  $\times$  age-group interaction leaves the estimate essentially unchanged (3.205\*\*, SE 1.055); reweighting to balance pre-period age groups attenuates it by about 19% to 2.597\* (SE 1.050), still significant at the 5% level but below the design’s minimum detectable effect, the largest single move any check produces.

**Sample.** Excluding the 2023 survey year, whose final quarter includes the start of the October 2023 war (Section 8), the triple interaction is 2.979\* (SE 1.170), smaller because 2023 carries the largest event-study coefficient but still significant at the 5% level, while the plain hours DiD stays a null (0.071, SE 0.198). Removing each of the forty occupations in turn (Appendix Figure A1) leaves the estimate between 2.415, without science and engineering professionals, and 3.580, without business and administration associate professionals; the largest move is 0.79 of the headline standard error, no refit leaves the band of one standard error around the headline, and every refit is significant at the 5% level on its own analytic standard error ( $p \leq 0.017$ ). Neither the six top-quartile occupations nor

the largest cells carry the result on their own.

**Outcome coding.** The hours variable is a bin-median approximation and the effect is smaller than one bin width, so the result should not depend on how the bins are turned into hours. Dropping the two imputed irregular-hours codes gives 3.056\*\* (SE 1.059). Because 35 and 40 hours are exact bin edges, the same specification can be estimated as a linear probability model: the probability of usually working at least 35 hours rises by 0.096\*\* (SE 0.029) per unit of exposure for mothers relative to childless women after 2021, and of at least 40 hours by 0.107\*\* (SE 0.038), or 1.9 and 2.1 percentage points per standard deviation against pre-period shares of 71% and 61%. The effect is a shift of mothers in teleworkable jobs across the full-time boundary.

**Inference.** Clustering two ways, by individual and by occupation, leaves the headline standard error unchanged at 1.022. The two procedures that address the forty-cluster problem disagree with the analytic  $p$ -value of 0.003: the wild cluster bootstrap gives  $p = 0.056$ , with a bootstrap 95% confidence set of  $[-0.26, 5.84]$ , and the permutation test gives  $p = 0.032$ , with 31 of 999 reassignments of the forty scores producing a  $|t|$  at least as large as the observed 3.15. The estimate is significant at the 1%, 5% and 10% levels under the three procedures, and we report all three rather than choosing among them; the bootstrap, which re-signs whole-cluster residuals, is the most conservative, while the permutation test holds the cluster structure fixed and randomizes only the regressor, the sharper question for a design whose identifying variation is occupation-level exposure.

## 6.5 Falsification: permuting exposure across occupations

Figure 5 plots the reference distribution the permutation test builds. Each of the 999 draws reassigns the forty calibrated scores at random across the forty occupation codes, leaving every woman’s occupation, mother status and survey year untouched, and refits equation (2). Under this sharp null the central 95% of cluster-robust  $t$ -statistics lie between  $-2.96$  and  $2.60$ ; the observed 3.15 is exceeded in absolute value by 31 draws, so  $p = 0.032$  (Monte-Carlo standard error 0.006). A design with this cluster structure produces an exposure gradient as large as ours about three times in a hundred when exposure carries no information, which is the falsification a placebo should supply and a fathers’ regression, a different and also-treated population, cannot.

## 6.6 Heterogeneity by the age of the youngest child

If remote work relaxes a time constraint that binds through child care, the response should be largest for mothers of young children and smallest for mothers of teenagers. Table 5 and Figure 6 re-estimate the hours DiD and DDD four times, each comparing the mothers

Table 4: Hours DDD: sensitivity of Mother  $\times$  Post  $\times$  WFH\_Exposure

| Specification                                           | Coefficient    | SE    | $p$              | $N$     | Clusters         |
|---------------------------------------------------------|----------------|-------|------------------|---------|------------------|
| <i>Exposure measure</i>                                 |                |       |                  |         |                  |
| Calibrated (primary)                                    | 3.224**        | 1.022 | 0.0031           | 246,326 | 40               |
| External (Dingel–Neiman)                                | 0.672          | 0.884 | 0.4517           | 246,326 | 40               |
| Realized (2021 anchor)                                  | 5.195*         | 2.363 | 0.0346           | 246,037 | 36               |
| Calibrated on men only (12 of 40 swapped)               | 3.869***       | 0.979 | 0.0003           | 246,326 | 40               |
| External index with swapped-occupation terms (40 occ.)  | 2.339          | 1.352 | 0.0915           | 246,326 | 40               |
| Mother $\times$ Post $\times$ Swapped (same model)      | −1.347         | 0.926 | 0.1537           | 246,326 | 40               |
| <i>Age balance</i>                                      |                |       |                  |         |                  |
| Age-interacted (+ Mother $\times$ age group)            | 3.205**        | 1.055 | 0.0042           | 246,326 | 40               |
| Reweightd (pre-period age-group raking)                 | 2.597*         | 1.050 | 0.0178           | 242,552 | 40               |
| <i>Sample</i>                                           |                |       |                  |         |                  |
| Unswapped occupations only                              | 4.309**        | 1.192 | 0.0011           | 132,046 | 30               |
| Excluding survey year 2023                              | 2.979*         | 1.170 | 0.0151           | 208,833 | 39               |
| Leave-one-occupation-out (40 refits): range             | [2.415, 3.580] |       | [0.0007, 0.0167] |         | 39               |
| <i>Occupational sorting</i>                             |                |       |                  |         |                  |
| Pre-period cell exposure as regressor, per SD           | 0.139          | 0.159 | 0.3823           | 247,931 | 245              |
| Exposure score as outcome: Mother $\times$ Post         | 0.001          | 0.003 | 0.6996           | 246,326 | individual       |
| Top-quartile indicator as outcome: Mother $\times$ Post | 0.004          | 0.007 | 0.5433           | 246,326 | individual       |
| <i>Outcome coding</i>                                   |                |       |                  |         |                  |
| Irregular-hours codes dropped, not imputed              | 3.056**        | 1.059 | 0.0064           | 232,785 | 40               |
| Full-time indicator ( $\geq 35$ hours)                  | 0.096**        | 0.029 | 0.0020           | 246,326 | 40               |
| Long-hours indicator ( $\geq 40$ hours)                 | 0.107**        | 0.038 | 0.0075           | 246,326 | 40               |
| <i>Inference on the primary estimate</i>                |                |       |                  |         |                  |
| Two-way clustering (individual, occupation)             | 3.224**        | 1.022 | 0.0031           | 246,326 | 40 $\times$ ind. |
| Wild cluster bootstrap ( $B = 9,999$ )                  | 3.224          | 1.022 | 0.0558           | 246,326 | 40               |
| Permutation test (999 reassignments)                    | 3.224          |       | 0.0320           | 246,326 | 40               |

Unless a row says otherwise: the triple interaction Mother  $\times$  Post  $\times$  WFH\_Exposure from equation (2), women working in the reference week, standard controls, calibrated exposure, clustered by occupation with the cluster count in the last column, and  $p$  the analytic  $t(G - 1)$   $p$ -value. The men-only row calibrates the index on men’s realized 2022–23 WFH shares. The swapped-occupation rows keep the external index as regressor on all forty occupations and add a Swapped indicator for the ten calibrated occupations with the full Mother  $\times$  Post structure; both triple interactions from that one model are shown. The unswapped-occupations row drops those ten. The leave-one-occupation-out row gives the range of the forty refits with one occupation removed (Appendix Figure A1) and the range of their  $p$ -values. Under *Occupational sorting*, the first row is the hours DDD with the pre-period demographic-cell index of Section 3.2 as regressor, reported per standard deviation of that index and clustered on the demographic cell; the next two are differences-in-differences with the occupation-level score, and an indicator for its top quartile, as the dependent variable, clustered by individual. The two indicator outcomes under *Outcome coding* are linear probability models on bin-crossing outcomes. The wild-bootstrap row reports the analytic coefficient and standard error with the bootstrap  $p$ -value; the permutation row reports the permutation  $p$ -value, which has no standard error.

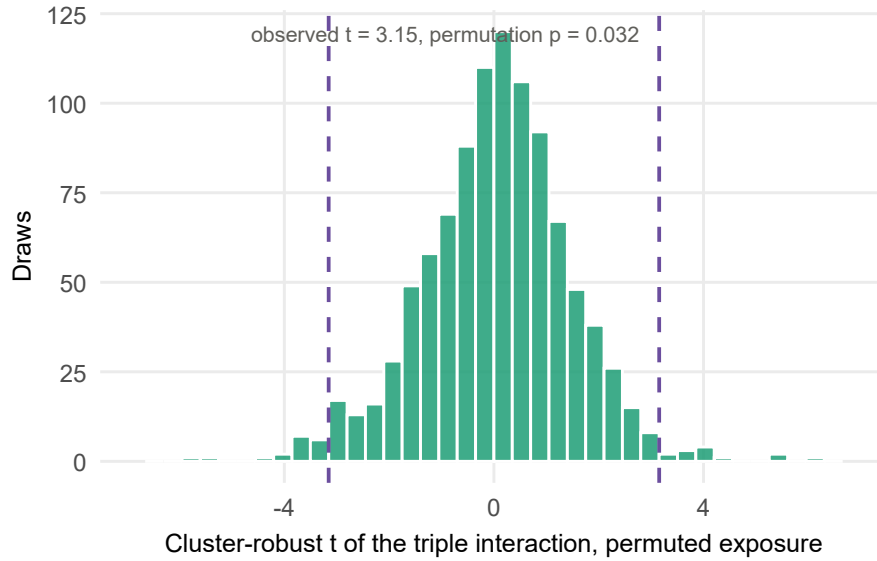


Figure 5: Permutation distribution of the hours DDD triple interaction  
*Notes:* Histogram of the cluster-robust  $t$ -statistic of Mother  $\times$  Post  $\times$  WFH\_Exposure over 999 random reassignments of the forty exposure scores across occupations, with the same sample, controls and clustering as column (2) of Table 2. Dashed lines mark  $\pm$  the observed statistic (3.15);  $p = 0.032$  counts the observed draw (Phipson and Smyth, 2010).

whose youngest child is aged 0–4, 5–9, 10–14 or 15–17 with the same childless women. The average effect is a null in every group. The exposure gradient declines monotonically with the child’s age: 3.656\*\* hours per unit of exposure (SE 1.079) for mothers of under-fives, 3.211\* (SE 1.311) for 5–9, 2.741 (SE 1.525) for 10–14 and 1.862 (SE 0.935) for 15–17, half the under-five estimate. The four regressions share the control group, so they are not independent samples and their intervals overlap; the pattern is a gradient consistent with the mechanism rather than a set of statistically separated effects, and it shows the estimate is not carried by mothers of teenagers, for whom the time constraint is weakest.

Table 5: Hours DiD and DDD by age of the youngest child

| Sample of mothers         | DiD: Mother $\times$ Post |       |         | DDD: Mother $\times$ Post $\times$ WFH_Exposure |       |         |
|---------------------------|---------------------------|-------|---------|-------------------------------------------------|-------|---------|
|                           | Coef.                     | SE    | $N$     | Coef.                                           | SE    | $N$     |
| All mothers (Table 2)     | 0.228                     | 0.181 | 251,857 | 3.224**                                         | 1.022 | 246,326 |
| Youngest child aged 0–4   | 0.127                     | 0.211 | 158,424 | 3.656**                                         | 1.079 | 154,499 |
| Youngest child aged 5–9   | 0.162                     | 0.253 | 130,782 | 3.211*                                          | 1.311 | 127,419 |
| Youngest child aged 10–14 | 0.159                     | 0.278 | 123,280 | 2.741*                                          | 1.525 | 120,084 |
| Youngest child aged 15–17 | 0.561                     | 0.348 | 107,061 | 1.862*                                          | 0.935 | 104,079 |

Weekly hours, reference-week workers, standard controls; each row compares the named group of mothers with all childless women, so the rows are not independent samples. DiD clustered by individual; DDD clustered by occupation, calibrated exposure. The first row repeats Table 2. No coefficient is dropped for collinearity in any column.

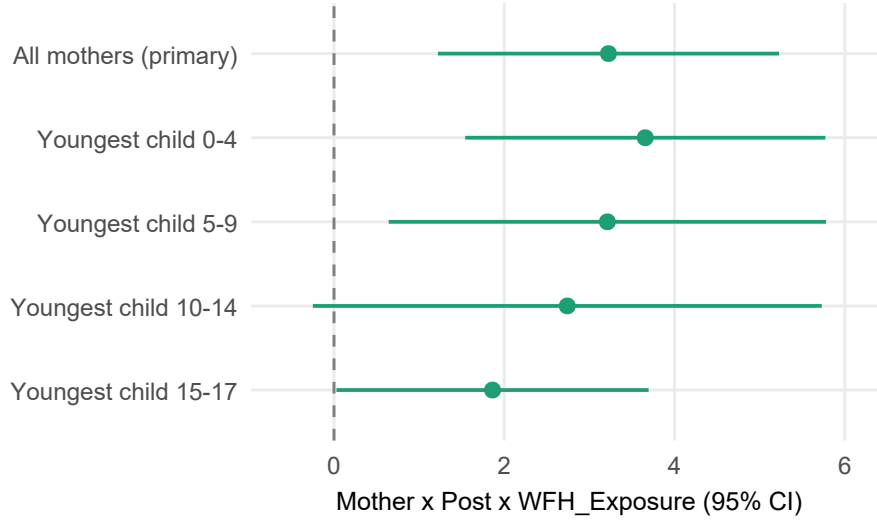


Figure 6: Hours DDD triple interaction by age of the youngest child  
*Notes:* The DDD column of Table 5, with 95% confidence intervals clustered by occupation; the top row is the pooled estimate of Table 2.

## 6.7 Heterogeneity by population group, and the fathers' comparison

Table 6 re-estimates the hours DiD and DDD for Jewish and Arab women and for men. For Jewish women the pattern is the pooled pattern: a DiD of 0.302 (SE 0.203) and a DDD of 2.847\*\* (SE 0.913). For Arab women the DiD is  $-0.072$  (SE 0.479) and the DDD 5.507\* (SE 2.674), nominally twice the Jewish estimate but statistically indistinguishable from it ( $z = 0.942$ ,  $p = 0.35$ ) and resting on a subsample nine times smaller and concentrated in few occupations; it is imprecise rather than a documented heterogeneous effect.

The fathers' comparison is a comparison, not a placebo. Re-estimating the same models on men, with “mother” read as “father,” gives a DiD of  $-0.400^*$  (SE 0.188) and a DDD of  $-1.762$  (SE 1.015,  $p = 0.09$ ). Fathers in teleworkable occupations are exposed to remote work exactly as mothers are, so a null was never the prediction; what the comparison shows is that the exposure gradient has the opposite sign for fathers, and that the two gradients are distinguishable ( $z = 3.461$ ,  $p = 0.0005$ ). A macro trend common to all workers in teleworkable jobs cannot produce a positive gradient for mothers beside a negative one for fathers, and a negative gradient for fathers is what within-household reallocation would look like if mothers' additional paid hours were matched by fathers' additional care hours. We report that as a reading the data are consistent with, not as a result the design identifies.

Table 6: Hours DiD and DDD by subgroup, and the fathers’ comparison

| Subgroup                     | DiD: Mother $\times$ Post |       |         | DDD: Mother $\times$ Post $\times$ WFH_Exposure |       |         |
|------------------------------|---------------------------|-------|---------|-------------------------------------------------|-------|---------|
|                              | Coef.                     | SE    | $N$     | Coef.                                           | SE    | $N$     |
| All women (primary)          | 0.228                     | 0.181 | 251,857 | 3.224**                                         | 1.022 | 246,326 |
| Jewish women                 | 0.302                     | 0.203 | 214,449 | 2.847**                                         | 0.913 | 209,727 |
| Arab women                   | -0.072                    | 0.479 | 24,975  | 5.507*                                          | 2.674 | 24,307  |
| Men (fathers vs. childless)  | -0.400*                   | 0.188 | 261,543 | -1.762*                                         | 1.015 | 251,779 |
| 95% CI, Jewish               | [-0.095, 0.699]           |       |         | [1.058, 4.635]                                  |       |         |
| 95% CI, Arab                 | [-1.010, 0.866]           |       |         | [0.266, 10.748]                                 |       |         |
| Jewish vs. Arab, $z$ ( $p$ ) | 0.720 (0.471)             |       |         | -0.942 (0.346)                                  |       |         |
| Women vs. men, $z$ ( $p$ )   | 2.406 (0.016)             |       |         | 3.461 (0.001)                                   |       |         |

Weekly hours, reference-week workers, standard controls. DiD clustered by individual; DDD clustered by occupation. Men: “Mother” read as “Father,” a comparison population rather than a placebo (Section 5.6). Two-sample  $z$ -tests as described there. Confidence intervals use the normal critical value. The Arab-women DiD drops the religion category ‘other’, constant on that sample; the Arab-women DDD drops the religion category ‘other’, constant on that sample; no other column drops a coefficient.

## 6.8 The extensive margin, briefly

Column (1) of Table 7 reports equation (1) with employment as the outcome: Mother  $\times$  Post = -0.0053 (SE 0.0052). The mother/non-mother employment gap did not change after the shift to remote work, and with a standard error of half a percentage point this is an informative null; the pre-period event-study coefficients pass their joint test ( $F(2, 79,069) = 0.63$ ,  $p = 0.53$ ). Column (2) reports the employment DDD with the cell-based index: -0.0257 (SE 0.0725). This null is of a different kind. The design could detect a triple interaction no smaller than 1.43 percentage points per standard deviation of the cell index, or 1.85% of the baseline employment rate, and against Harrington and Kahn’s 0.78 percentage points per 10% rise in WFH it is underpowered by a factor of two to three. The shortfall is structural, a binary outcome and a regressor with a standard deviation of 0.07, and re-clustering, combining the two margins and refining the cell partition did not close it. The employment DDD is therefore *uninformative*, evidence neither for nor against a WFH-mediated employment effect, and the two nulls in Table 7 should not be read as one verdict. This is what motivates the intensive-margin design, where occupation-level exposure is available.

## 7 Discussion

Among mothers who remain in work there is no post-2021 hours effect on average, and a large one in teleworkable occupations. The conjunction is the paper’s result: a triple interaction of about 3.2 hours per unit of exposure, an implied change at zero exposure that is if anything negative, and a selection correction the estimate survives. This is

Table 7: Extensive margin: employment DiD and DDD

|                                               | (1) DiD               | (2) DDD, cell-based exposure       |
|-----------------------------------------------|-----------------------|------------------------------------|
| Mother $\times$ Post $\times$ WFH_Exposure    |                       | −0.0257<br>(0.0725)                |
| Mother $\times$ Post                          | −0.0053<br>(0.0052)   | 0.0065<br>(0.0168)                 |
| Mother $\times$ WFH_Exposure                  |                       | −0.1404<br>(0.0803)                |
| Post $\times$ WFH_Exposure                    |                       | −0.1555**<br>(0.0566)              |
| WFH_Exposure                                  |                       | 0.0131<br>(0.0565)                 |
| Mother                                        | −0.0034<br>(0.0041)   |                                    |
| Post                                          | 0.0153***<br>(0.0043) |                                    |
| Constant                                      | 0.4926***<br>(0.0085) |                                    |
| MDE ( $\alpha = 0.05$ , power 0.80), per unit |                       | 0.2031                             |
| MDE per SD of exposure                        |                       | 0.0143                             |
| as % of baseline employment rate              |                       | 1.85%                              |
| Observations                                  | 364,784               | 355,984                            |
| $R^2$                                         | 0.227                 | 0.221                              |
| Clustering                                    | individual            | demographic cell ( $\approx 210$ ) |

Dependent variable: employment indicator; full sample of employed and non-employed women; standard controls in both columns, four decimals because the outcome is a probability. Column (2) adds Mother  $\times$  age-group interactions, so its Mother and Post main effects are not separately reported, and uses the demographic-cell index of Section 3.2, whose standard deviation on this sample is 0.0706; the per-SD MDE row is the interpretable one. Baseline employment rate 0.7737. No coefficient is dropped for collinearity in any column.

consistent with WFH access relaxing a binding time constraint for mothers in teleworkable jobs, the [Goldin \(2014\)](#) mechanism, and hard to reconcile with a general recovery common to all women, which would have raised hours across occupations rather than only where the work can be done from home.

How precisely the effect is estimated is a separate question from whether it is there. The three inference procedures of Section 6.4 place the estimate at the 1%, 5% and 10% levels, and what makes the finding credible is the pattern of evidence rather than any one  $p$ -value: a saturated specification that changes nothing, a top-quartile contrast significant on its own, a leave-one-out range that never leaves the headline’s standard error, a monotone child-age gradient, an opposite-signed fathers’ gradient and a selection correction that barely moves the bounds. Which occupations do the work is concrete: the calibration moved teaching and clerical support sharply down, because Israeli practice kept them in person regardless of task content, while ICT professionals stayed high on both measures, and the gradient is identified off that contrast. The checks that calibrate on men only or hold the sample fixed while letting the swapped occupations differ show the calibration is not what produces the result.

The magnitude is economically meaningful where it applies. For a mother at the top quartile’s mean exposure the regression implies a post-2021 change of about 1.4 hours a week ( $-0.54 + 3.22 \times 0.59$ ), and the raw top-quartile difference-in-differences of Figure 2 is 1.55. The penalty it is set against must be the penalty in those same jobs, not the average one: column (2) of Table 2 puts the pre-period motherhood gap at  $-1.45 - 1.30 \times 0.59 \approx -2.2$  hours at that exposure, so mothers in the most teleworkable jobs recovered roughly three fifths of their penalty after 2021. At the bottom quartile’s mean exposure of 0.06 the implied change is  $-0.35$  hours: mothers in the least teleworkable jobs recovered none of it. The size is what the most literal form of the time-constraint mechanism would produce: [Aksoy et al. \(2023\)](#) put the commuting time saved at about 72 minutes per day worked from home, so two remote days a week free roughly two and a half hours, and a mother who returns most of that time to paid work is a mother whose hours rise by about the amount estimated here.

A general recovery is not the only alternative to that mechanism; three demand-side accounts also fit a gradient by teleworkability. Employers in professional and ICT occupations may have changed the contracts they offer mothers after 2021, with no change in mothers’ constraints; the saturated specification absorbs any occupation-by-year shock common to mothers and childless women, so this needs a change aimed at mothers in teleworkable occupations specifically. The 2021–22 hiring boom in the technology sector raised hours in exposed occupations for everyone; the fathers’ opposite-signed gradient rules out a shock common to all workers in those occupations, but not one that fell on mothers alone. And usual hours are self-reported: if mothers working from home report the hours they are available while fathers report the hours they work, the two gradients

would diverge as observed with no change in labor supplied; the child-age gradient weighs against this, since a reporting difference has no reason to be twice as large for mothers of infants as for mothers of teenagers while a binding time constraint does. None of the three is excluded by the design, and we read the exposure gradient as the time-constraint mechanism’s signature rather than its proof.

The timing is also informative. The DDD event study rises from 3.0 in 2021 to 4.0 in 2023, consistent with WFH adoption and household time reallocation being a multi-year adjustment; the 2023 coefficient is the largest on both margins and, for the reason given in Section 8, should be held more loosely than the others.

Why hours and not employment? The employment DDD is not a failed test of the mechanism but an uninformative one, underpowered by a factor of two to three against the closest published effect. Substantively the two results are coherent: the effect operates on *how much* mothers in teleworkable jobs work, not on *whether* they are employed, which is the margin Kleven et al. (2019b) identify as a leading channel of the child penalty. This is where our findings depart from Harrington and Kahn (2025), whose US evidence is on employment and income. The policy reading is correspondingly narrow: WFH availability matters for how intensively mothers who already hold *teleworkable* jobs can work, not for whether mothers in general hold jobs. The one heterogeneity we read substantively is by the age of the youngest child, because the mechanism predicts its direction in advance; the Arab-women estimate is larger than the Jewish one but imprecise and statistically indistinguishable from it, and we do not claim heterogeneity on its basis.

## 8 Limitations

**Identification.** The triple difference’s own pre-trend test (Section 6.3) shows no visible violation but lacks the power to rule one out; the exposure measure that delivers the headline draws on post-period Israeli data for its calibration, and although the checks of Section 6.4 show the calibration does not produce the result, the fully pre-treatment external index yields a positive and insignificant estimate; the crosswalk behind that external index is undocumented (Section 3.2); and the sorting check finds no post-2021 movement of mothers across exposure levels but bounds the effect only imprecisely on a regressor fixed before treatment. The three inference procedures place the headline at the 1%, 5% and 10% levels, and that range, not the analytic  $p$ -value, is the paper’s statement of how precisely the effect is estimated.

**The narrowed hours estimand.** Hours are usual hours among women who were working, not among all women holding a job, and roughly 10% of each year’s employed sample is excluded on that basis. The exclusion is not innocuous for the triple interaction because absence is related to exposure: it runs at 14.4% of the employed in the least teleworkable

quartile against 8.5–9.6% in the other three, and among mothers it fell after 2021 in the top quartile while rising in the bottom one.<sup>1</sup> Movement across the absence margin is part of what selection into the observed-hours sample means here, which is why the bounds of Section 5.5 define selection on that sample. They handle only *excess* selection in the post-period mother cell, assume monotone selection to justify trimming from a single tail, and found little to correct; that is not the same as the concern being absent.

**Survey weights.** CBS survey weights are intentionally not applied in any outcome regression; reported rates and coefficients are unweighted estimates on the analysis sample, not population-representative statistics. The one exception is internal to constructing the cell-based exposure index.

**The 2023 survey year.** The 2023 coefficient is the largest in both event studies, and the 2023 survey year includes the quarter in which the war that began on 7 October 2023 disrupted the labor market: reserve call-ups, which bear on the fathers’ comparison, and school closures and evacuations, which bear on mothers’ hours. The estimate survives dropping the year and 2021 and 2022 are individually significant on the triple difference, but the 2023 point should be held more loosely. Early 2021 likewise included a third lockdown, so the post-period is “post-acute” more cleanly in its later years than in its first.

**Who is childless, and age.** The control group is any woman aged 25–59 with no child under 17 in the household, so it includes women whose children are grown and women who become mothers within the window, as in [Harrington and Kahn \(2025\)](#); if part of it responds to the treatment as mothers do, the DiD is biased toward zero. Age enters as a categorical code because the extract has no continuous age, and mothers and non-mothers are imbalanced on it in a way that tracks exposure (Table A2); the age-interacted and reweighted specifications address this and the estimate survives, but the reweighted attenuation ( $3.224 \rightarrow 2.597$ ) is the largest single move any check produces.

**Exposure is measured coarsely, and on a mixed scale.** Exposure is assigned at the two-digit occupation level, forty groups, and [Buzaglo-Baris \(2023\)](#) shows that within-firm sorting matters as much as occupation for Israeli gender gaps. The calibrated score mixes task-based probabilities for thirty occupations with realized shares for ten, which costs cross-sectional fit; a score on a single scale is the realized index of Table 4, which gives a larger and noisier estimate. Within-occupation heterogeneity in teleworkability is classical

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<sup>1</sup>Reference-week absence among employed women with a matched occupation, by the quartiles of Figure 2: 14.4%, 8.5%, 9.5% and 9.6%. Among mothers the top-quartile share fell from 12.1% to 10.4% after 2021 and the bottom-quartile share rose from 15.5% to 16.9%; among childless women the top-quartile share is 6.1% in both periods.

measurement error in the regressor and attenuates the triple interaction toward zero. Occupation codes masked by the CBS for disclosure control are dropped when building exposure; a proxy check finds no strong evidence of bias from this. The panel does not support within-person identification: 1.9% of women span the pre/post boundary.

## 9 Conclusion

**[SKELETON ONLY — REPLACE WITH YOUR OWN TEXT BEFORE SUBMISSION]**

1. **Paragraph 1 — the question and the move.** *Harrington and Kahn (2025) for the US; this paper takes it to Israel and to hours conditional on working. Not a replication (Sections 2, 3).*
2. **Paragraph 2 — the result as a conjunction.** *No average effect: the hours DiD is 0.228 (SE 0.181), a reasonably precise null that the selection correction widens rather than overturns. An effect confined to teleworkable work: the triple interaction is 3.224 (SE 1.022,  $p < 0.01$ ), survives the generalized correction (Imbens–Manski [1.021, 5.324]) and two age-balance specifications, and at zero exposure the implied change is negative. Fathers show no exposure gradient, and the women/men difference is  $z = 3.461$  (Sections 6.1–6.7).*
3. **Paragraph 3 — what it means.** *Goldin’s time-constraint mechanism: the response is where the work can be done from home, and nowhere else. A result about which mothers benefit, not about mothers as a group. Measurement is part of it: the mechanism is visible only once exposure is measured as Israeli jobs were actually done (Section 7).*
4. **Paragraph 4 — what the design cannot say.** *Employment: the DiD is a precise null, the DDD is uninformative (MDE 1.43 percentage points per SD of exposure, two to three times the Harrington benchmark), and the two should not be conflated. Earnings: not observed. The calibrated exposure is partly post-treatment; the 2023 year includes the war; inference rests on forty clusters; estimates are unweighted (Sections 6.8, 8).*
5. **Paragraph 5 — policy reading and next steps.** *WFH availability bears on how intensively mothers already employed in teleworkable jobs work, not on whether mothers are employed. Next: an earnings outcome, which needs income data linked to the survey since the extract carries none; a higher-powered test of the employment margin; a jointly estimated model letting the mechanism differ by sex and population group.*

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## A Appendix

Table A1 lists, for each of the forty two-digit ISCO-08 occupations, the external [Dingel and Neiman \(2020\)](#) score, the realized share usually working from home in 2022–23 that the calibration tests it against, whether the score was swapped, the calibrated score the hours DDD uses, and the realized reference-week share over 2021–23 on men and women pooled. Figure A2 plots the last against the calibrated score: the ten swapped occupations sit above the 45-degree line, because their calibrated value is the usual-place share, which is lower than the reference-week share everywhere, and the thirty unswapped ones lie below it, the visual form of the scale-mixing Section 3.2 describes. Table A2 reports the pre-period covariate balance between mothers and childless women within each exposure quartile, and Figure A1 the leave-one-occupation-out refits of the headline estimate.

Table A1: External, realized and calibrated WFH exposure by two-digit occupation

| ISCO | Occupation (ISCO-08 sub-major group)                           | Ext.  | Real. 22–23 | Swap | Calib. | Ref.-wk 21–23 | <i>N</i> |
|------|----------------------------------------------------------------|-------|-------------|------|--------|---------------|----------|
| 11   | Chief executives and legislators                               | 0.900 | 0.155       | yes  | 0.155  | 0.320         | 2,855    |
| 12   | Administrative and commercial managers                         | 0.892 | 0.143       | yes  | 0.143  | 0.336         | 3,859    |
| 13   | Production and specialised services managers                   | 0.638 | 0.097       | yes  | 0.097  | 0.199         | 9,990    |
| 14   | Hospitality, retail and other services managers                | 0.773 | 0.066       | yes  | 0.066  | 0.132         | 2,827    |
| 21   | Science and engineering professionals                          | 0.644 | 0.210       |      | 0.644  | 0.349         | 13,228   |
| 22   | Health professionals                                           | 0.101 | 0.059       |      | 0.101  | 0.088         | 9,665    |
| 23   | Teaching professionals                                         | 0.966 | 0.069       | yes  | 0.069  | 0.184         | 20,496   |
| 24   | Business and administration professionals                      | 0.943 | 0.233       | yes  | 0.233  | 0.449         | 10,479   |
| 25   | ICT professionals                                              | 1.000 | 0.360       | yes  | 0.360  | 0.615         | 14,638   |
| 26   | Legal, social and cultural professionals                       | 0.678 | 0.215       |      | 0.678  | 0.371         | 13,344   |
| 31   | Science and engineering associate professionals                | 0.144 | 0.059       |      | 0.144  | 0.106         | 3,926    |
| 32   | Health associate professionals                                 | 0.057 | 0.080       |      | 0.057  | 0.084         | 2,203    |
| 33   | Business and administration associate professionals            | 0.684 | 0.123       | yes  | 0.123  | 0.223         | 22,415   |
| 34   | Legal, social and cultural associate professionals             | 0.577 | 0.142       |      | 0.577  | 0.206         | 5,480    |
| 35   | ICT technicians                                                | 0.750 | 0.244       |      | 0.750  | 0.380         | 2,491    |
| 41   | General and keyboard clerks                                    | 1.000 | 0.098       | yes  | 0.098  | 0.144         | 4,846    |
| 42   | Customer services clerks                                       | 0.292 | 0.096       |      | 0.292  | 0.159         | 3,875    |
| 43   | Numerical and material recording clerks                        | 0.526 | 0.071       |      | 0.526  | 0.152         | 4,948    |
| 44   | Other clerical support workers                                 | 0.625 | 0.083       | yes  | 0.083  | 0.161         | 1,049    |
| 51   | Personal service workers                                       | 0.271 | 0.146       |      | 0.271  | 0.144         | 7,808    |
| 52   | Sales workers                                                  | 0.200 | 0.050       |      | 0.200  | 0.063         | 8,704    |
| 53   | Personal care workers                                          | 0.300 | 0.046       |      | 0.300  | 0.045         | 17,206   |
| 54   | Protective services workers                                    | 0.100 | 0.008       |      | 0.100  | 0.011         | 3,969    |
| 61   | Market-oriented skilled agricultural workers                   | 0.182 | 0.027       |      | 0.182  | 0.051         | 1,325    |
| 62   | Market-oriented skilled forestry, fishery and hunting workers  | 0.179 | 0.000       |      | 0.179  | 0.045         | 22       |
| 63   | Subsistence farmers, fishers, hunters and gatherers            | 0.000 | 0.375       |      | 0.000  | 0.375         | 8        |
| 71   | Building and related trades workers                            | 0.027 | 0.008       |      | 0.027  | 0.016         | 5,208    |
| 72   | Metal, machinery and related trades workers                    | 0.000 | 0.008       |      | 0.000  | 0.013         | 3,982    |
| 73   | Handicraft and printing workers                                | 0.138 | 0.118       |      | 0.138  | 0.129         | 666      |
| 74   | Electrical and electronic trades workers                       | 0.000 | 0.022       |      | 0.000  | 0.042         | 3,424    |
| 75   | Food, wood, garment and other craft workers                    | 0.073 | 0.122       |      | 0.073  | 0.109         | 2,304    |
| 81   | Stationary plant and machine operators                         | 0.000 | 0.019       |      | 0.000  | 0.017         | 2,407    |
| 82   | Assemblers                                                     | 0.000 | 0.012       |      | 0.000  | 0.015         | 1,166    |
| 83   | Drivers and mobile plant operators                             | 0.237 | 0.006       |      | 0.237  | 0.006         | 8,480    |
| 91   | Cleaners and helpers                                           | 0.000 | 0.004       |      | 0.000  | 0.003         | 3,474    |
| 92   | Agricultural, forestry and fishery labourers                   | 0.000 | 0.007       |      | 0.000  | 0.016         | 192      |
| 93   | Labourers in mining, construction, manufacturing and transport | 0.120 | 0.006       |      | 0.120  | 0.008         | 2,659    |
| 94   | Food preparation assistants                                    | 0.000 | 0.013       |      | 0.000  | 0.011         | 649      |
| 95   | Street and related sales and service workers                   | 0.000 | 0.000       |      | 0.000  | 0.043         | 23       |
| 96   | Refuse workers and other elementary workers                    | 0.250 | 0.022       |      | 0.250  | 0.024         | 493      |

Ext.: the [Dingel and Neiman \(2020\)](#) teleworkability score at the two-digit ISCO-08 level, via the crosswalk described in Section 3.2. Real. 22–23: share of employed respondents (men and women, all ages in the extract) who usually work from home in 2022–23, the calibration’s benchmark; Swap: the calibration replaced the external score with it. Calib.: the score the hours DDD uses. Ref.-wk 21–23: share who worked from home in the survey reference week over 2021–23, men and women pooled, with *N* the pooled employed sample carrying that item. Codes 1–3 (armed forces) carry no external score and are not in the sample.

Table A2: Pre-period covariate balance by quartile of occupation-level WFH exposure

| Mothers minus childless women, pre-period | Q1                 | Q2                 | Q3                 | Q4               |
|-------------------------------------------|--------------------|--------------------|--------------------|------------------|
| Age-group code (CBS 3–7), mean            | −0.54***<br>(0.03) | −0.44***<br>(0.03) | −0.33***<br>(0.03) | 0.08*<br>(0.04)  |
| Married (%)                               | 32.2***<br>(1.1)   | 33.7***<br>(1.1)   | 41.5***<br>(0.9)   | 43.1***<br>(1.3) |
| Academic degree (%)                       | 10.8***<br>(1.2)   | 4.5***<br>(1.2)    | 1.5<br>(1.0)       | 1.8<br>(1.2)     |
| Below high school (%)                     | −3.2***<br>(0.6)   | −1.7***<br>(0.4)   | −3.3***<br>(0.5)   | −0.2<br>(0.2)    |
| Jewish household head (%)                 | 1.6<br>(1.0)       | 0.5<br>(0.7)       | 14.3***<br>(0.9)   | 3.1***<br>(0.8)  |
| Center or Tel Aviv district (%)           | −4.4***<br>(1.2)   | −3.4**<br>(1.2)    | −8.8***<br>(1.1)   | −4.5**<br>(1.4)  |
| Arab (%)                                  | 1.9*<br>(0.9)      | 0.2<br>(0.6)       | 3.7***<br>(0.6)    | −0.9<br>(0.6)    |
| Mothers                                   | 22,976             | 24,664             | 22,562             | 16,377           |
| Childless women                           | 11,420             | 11,586             | 17,244             | 8,352            |
| Occupations                               | 15                 | 7                  | 11                 | 6                |
| Exposure range                            | 0.00–0.10          | 0.10–0.14          | 0.14–0.30          | 0.36–0.75        |

Women working in the reference week in 2017–2019 with a matched two-digit occupation, the hours DDD’s pre-period estimation sample, split into the exposure quartiles of Figure 2. Each cell is the mother-minus-childless difference in the row variable within the quartile, with a standard error clustered by individual in parentheses; shares are in percentage points, the age-group code in code units (CBS codes 3–7 span ages 25–59 in five-year bands). Stars follow the paper’s legend on a normal reference. Unweighted.



Figure A1: Leave-one-occupation-out: the hours DDD triple interaction with each occupation removed

*Notes:* Each point is  $\text{Mother} \times \text{Post} \times \text{WFH\_Exposure}$  from equation (2) re-estimated without the occupation named on the axis (column (2) of Table 2 on 39 occupations), with its 95% confidence interval clustered by occupation; rows are sorted by the estimate. The dashed line is the headline estimate on all forty occupations and the shaded band one headline standard error either side of it; the dotted line marks zero. Triangles would mark refits outside the band; there are none.

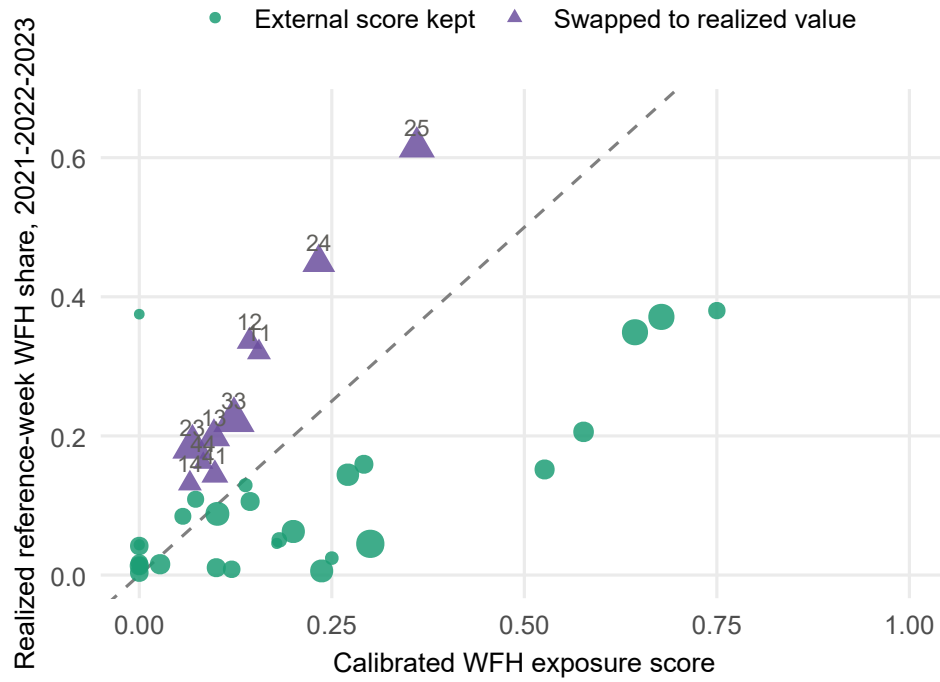


Figure A2: First stage: calibrated exposure score against realized reference-week WFH, by occupation

*Notes:* One point per two-digit occupation, sized by the pooled employed sample in 2021–23; triangles are the ten occupations the calibration swapped to their realized value, labelled by ISCO-08 code. The dashed line is the 45-degree line. Statistics in the text of Section 3.2.